AMRITA VISHWA VIDYAPEETHAM

Amrita School of Engineering, Chennai

Course Delivery Plan			
Course Code: Course Name	**19MEE347: Total Quality Management**	**Program**	**Artificial Intelligence Engineering**
L - T – P - C	**3 - 0 - 0 - 3**	**Semester / Year**	**VII / IV**
Name(s) of the Faculty	**Dr Sunil Kumar**	**Pre-requisite**	**Nil**
Course Mentor	**Dr Sunil Kumar**	**Academic Year**	**2026-27**
Course Overview	Total quality management (TQM) is a philosophy, methodology and system of tools aimed to create and maintain mechanism of organizations continuous improvement. It involves all departments and employees into improvement of processes and products. It helps to reduce costs and to meet and exceed needs and expectations of customers and other stakeholders of an organization. The objective of this course is to introduce the main TQM principles and various TQM tools and techniques to address the quality problems in an organization.		

Course Objective		Course Outcomes	
1	To impart knowledge on quality management principles, tools, techniques and quality standards for real life applications	CO1	Evaluate the principles of quality management and to explain how these principles can be applied within quality management systems
		CO2	Evaluate the performance measures using various quality and management tools.
		CO3	Apply the Quality Function Deployment, Taguchi principles, Total Productive Maintenance and Failure Mode and Effect Analysis concepts to solve industrial problems.
		CO4	Practice the various quality system in industry.

Syllabus

Unit 1

Definition of quality - dimensions of quality. Quality planning - quality costs. Total Quality Management: historical review and principles – leadership - quality council - quality statements - strategic planning - Deming philosophy. Barriers to TQM implementation.

Unit 2

Customer satisfaction – Customer retention - Employee involvement - Performance appraisal - Continuous process improvement - Supplier partnership - Performance measures. Seven tools of quality. Statistical fundamentals - Control Charts for variables and attributes - Process capability - Concept of six sigma - New seven management tools - Benchmarking.

Unit 3

Quality function deployment (QFD) - Taguchi quality loss function - Total Productive Maintenance (TPM) - FMEA. Need for quality systems - ISO 9000:2000 – Elements of quality systems (such as ISO 9000:2000). Implementation of quality system – documentation - quality auditing - QS 9000-ISO 14000.

TEXT BOOK

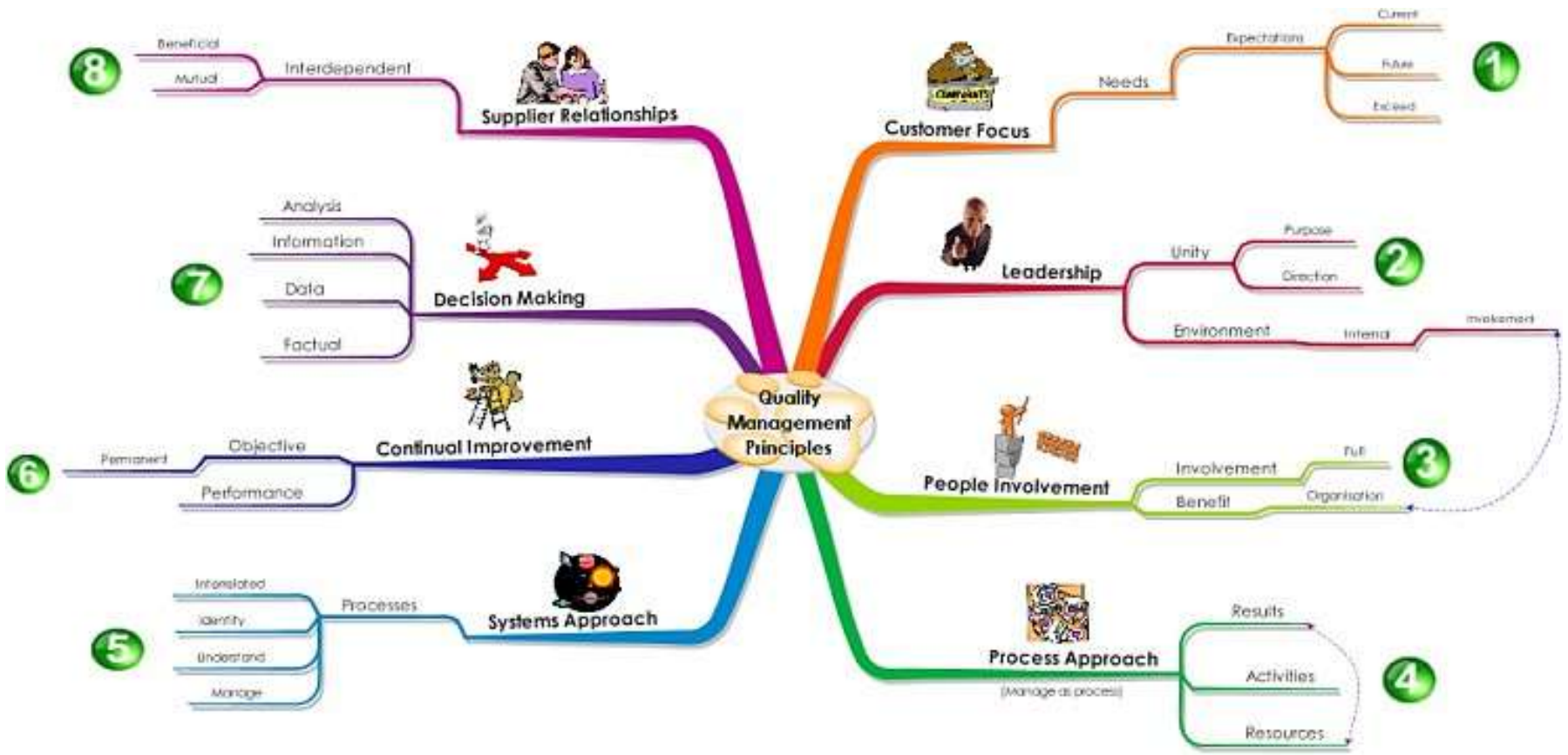
T1: Besterfield D. H. – “Total Quality Management” - Pearson Education Asia – 2015 - 4th Edition.

REFERENCES

R1: Evans J. R, and Lidsay W. M. – “The Management and Control of Quality” - Southwestern (Thomson Learning) - 2002 - 5th Edition.

R2: Feigenbaum A V – “Total Quality Management - Vol I & II” – McGraw Hill – 1991.

Concept Map


Quality Management Principles
Customer Focus
Needs
Expectations
Current
Future
Exceed
1
Leadership
Unity
Purpose
Direction
2
Environment
Internal
Involvement
People Involvement
Involvement
Full
Benefit
Organisation
3
Process Approach
(Manage as process)
Results
Activities
Resources
4
Systems Approach
Processes
Interrelated
Identify
Understand
Manage
5
Continual Improvement
Objective
Performance
Permanent
6
Decision Making
Analysis
Information
Data
Factual
7
Supplier Relationships
Interdependent
Beneficial
Mutual
8

<table>
<tr><th colspan="5">Evaluation and Grading</th></tr>
<tr><th colspan="3">Internal (70)</th><th>External (30)</th><th>Total (100)</th></tr>
<tr><td colspan="2">Components</td><td>Marks</td><td>Total Marks (70)</td><td rowspan="4">End Semester = 30</td><td rowspan="4">Internal + External = 100 marks</td></tr>
<tr><td colspan="2">Mid Term Examination</td><td>30</td><td>30</td></tr>
<tr><td rowspan="3">Continuous Assessments</td><td>Quiz (2)</td><td>20</td><td rowspan="2">40</td></tr>
<tr><td>Assignment – 1 (Essay type)
Assignment – 2 (Case Study and Problems)</td><td>10
10</td></tr>
<tr><th colspan="5">Programme Outcome (PO)</th></tr>
<tr><td>PO1</td><td colspan="4">Engineering knowledge: Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.</td></tr>
<tr><td>PO2</td><td colspan="4">Problem analysis: Identity, formulate, research literature, and analyze complex engineering problems reaching substantiated conclusions using the first principles of mathematics, natural sciences, and engineering sciences.</td></tr>
<tr><td>PO3</td><td colspan="4">Design/development of solutions: Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for public health and safety, and cultural, societal, and environmental considerations.</td></tr>
<tr><td>PO4</td><td colspan="4">Conduct investigations of complex problems: Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.</td></tr>
<tr><td>PO5</td><td colspan="4">Modern tool usage: Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.</td></tr>
<tr><td>PO6</td><td colspan="4">The engineer and society: Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.</td></tr>
<tr><td>PO7</td><td colspan="4">Environment and sustainability: Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.</td></tr>
<tr><td>PO8</td><td colspan="4">Ethics: Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.</td></tr>
<tr><td>PO9</td><td colspan="4">Individual and teamwork: Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.</td></tr>
</table>

PO10	**Communication:** Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.
PO11	**Project management and finance:** Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.
PO12	**Life-long learning:** Recognize the need for and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.
PSO1	Apply knowledge acquired in the field of Design, Manufacturing, Thermal, and Fluid sciences to solve real-world engineering problems using emerging technologies.
PSO2	Extend and implement innovative thinking on product design and development with the aid of modern CAD/CAM/CAE tools.
PSO3	Apply the Science and Engineering knowledge for materials design and processing for development and improvement of products and processes.

CO – PO Affinity Map

PO/PSO / CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO 1	1	2										2	2		
CO 2	1	2										2	2		
CO 3	2	2	2									2	2		
CO 4	2	2	2	2								2	2		

3 – Strong, 2 Moderate, 1 -weak

Threshold (%)	Target (%)	Percentage of Students		
		Level 1	Level 2	Level 3
50%	**70%**	**50%**	**60%**	**70%**

Class	Topics to be covered	Mode of Teaching	In-Class Activities	Out - Class Activities (E – References)	CO Mapping	Reference
Unit 1						
1	Introduction to Course, CDP, Evaluation Scheme, Evolution of Quality, Quality Definition	Smartboard, Presentations	Discussion about need for quality in organisation	https://www.intechopen.com/chapters/53946 https://asq.org/quality-resources/history-of-quality	1	R1
2	Introduction to Quality Gurus, Dimensions of Product and Service Quality	Smartboard, Presentations	Discussion about quality gurus who contributed much in quality establishment	https://www.referenceforbusiness.com/management/Pr-Sa/Quality-Gurus.html https://www.encyclopedia.com/management/encyclopedias-almanacs-transcripts-and-maps/quality-gurus	1	R1
3	Dimensions of Product and Service Quality	Smartboard, Presentations	Discussion about basic concepts in Service and Product quality	https://www.researchgate.net/publication/266549233_Customer's_Perceptions_and_the_Dimensions_of_Quality	1	R1
4	Quality Planning and Quality Costs	Smartboard, Presentations	Discussion about quality planning and cost	https://www.youtube.com/watch?v=iCgzbYi_Iw8	1	T1
5	Introduction to TQM, Basic Concepts of TQM	Smartboard, Presentations	Discussion on concepts in TQM	https://www.economicsdiscussion.net/quality-management/total-quality-management/32329 https://ebrary.net/3027/management/basic_concepts_tqm	1	T1
6	TQM leadership, Quality Statements	Smartboard, Presentations	Discussion on various quality statements and leadership	https://www.youtube.com/watch?v=-MffNTFlFjw	1	R1

7	Quality Statements, Strategic Quality Planning, Quality Council	Smartboard, Presentations	Presentation on Strategic Quality Planning and Quality Council	http://aiqmindia.blogspot.com/2011/01/strategic-quality-planning-and.html https://asq.org/quality-resources/quality-plans	1	T1
8	Strategic Quality Planning, Quality Council	Smartboard, Presentations	Presentation on Strategic Quality Planning and Quality Council	https://www.youtube.com/watch?v=xUZumaHziUQ	1	R1
9	Deming's 14 Points for Management	Smartboard, Presentations	Discussion on Deming's principle on Quality	https://www.smartersolutions.com/services/business-system-iee/demings-14-points-explained-implementation	2	R1
10	Deming's 14 Points for Management	Smartboard, Presentations	Discussion on Deming's principle on Quality	https://asq.org/quality-resources/total-quality-management/deming-points https://www.youtube.com/watch?v=-tGsteslTTQ	2	R1
11	Barriers to TQM implementation	Smartboard, Presentations	Discussion on barriers in TQM	https://www.youtube.com/watch?v=PLAJAdNF0Wg	2	T1
Quiz 1						
Unit 2						
12	Customer Satisfaction, Customer Perception of Quality, Identifying Customer Needs	Smartboard, Presentations	Discussion on Customer satisfaction and need	https://clootrack.com/knowledge_base/what-is-customer-perception/	2	T1
Assignment 1						
13	Kano Model of Customer Satisfaction, Customer Retention	Smartboard, Presentations	Discussion on Customer retention	https://www.isme.in/kano-model-of-customer-satisfaction-and-its-importance	2	R2

14	Concept of Employee Involvement	Smartboard, Presentations	Discussion on Employee involvement	https://www.youtube.com/watch?v=nfX5cwldA2M	2	R2
15	Procedure for Performance appraisal	Smartboard, Presentations	Discussion on performance appraisal procedure	https://www.youtube.com/watch?v=IGgOO2ZGpf0	2	R1
16	Quality steps in Continuous process improvement	Smartboard, Presentations	Discussion on quality steps for improvement	https://www.youtube.com/watch?v=gdnPYtAN1ks	2	R2
17	Supplier partnership to enhance quality standard	Smartboard, Presentations	Discussion on quality steps for improvement	https://www.brainkart.com/article/Supplier-Partnership_5258/	2	T1
18	Methodology to measure Performance	Smartboard, Presentations	Discussion on steps to improve performance	https://www.youtube.com/watch?v=XRg7wX2-VK8	2	R1
19	Seven tools of quality	Smartboard, Presentations	Discussion on 7 tools in quality	https://youtu.be/yuH35ottILU	2	R1
20	Statistical fundamentals for quality measurement	Smartboard, Presentations	Presentation on statistical method for quality	https://www.youtube.com/watch?v=qb3mvJ1gb9g	2	R2
21	Control Charts for variables	Smartboard, Presentations	Discussion on control charts and Problem solving	https://www.youtube.com/watch?v=DcP_3Aspu7Q	2	R2
22	Control Charts for variables	Smartboard, Presentations	Discussion on control charts and Problem solving	https://www.youtube.com/watch?v=Aj7lJLR-7b4	2	R2

23	Control Charts for attributes	Smartboard, Presentations	Discussion on control charts and Problem solving	https://youtu.be/lT0muPy7lFI	2	R2
24	Process capability to maintain quality standard	Smartboard, Presentations	Discussion to maintain quality standard	https://techqualitypedia.com/process-capability/	2	R2
25	Concept of six sigma	Smartboard, Presentations	Presentation on Six sigma	https://www.youtube.com/watch?v=uB0MZ0dKQUo	2	R2
26	Six Sigma - Case study	Smartboard, Presentations	Discussion on Six sigma case study	https://www.isixsigma.com/implementation/case-studies/	2	R1
27	New seven management tools	Smartboard, Presentations	Discussion on 7 management tools	https://www.qualitygurus.com/seven-management-and-planning-tools/	2	T1
28	New seven management tools	Smartboard, Presentations	Discussion on 7 management tools	https://www.qualitygurus.com/seven-management-and-planning-tools/	2	T1
29	Benchmarking	Smartboard, Presentations	Discussion on Benchmarking	https://youtu.be/r7qSCNV-8mM	2	R1
30	Benchmarking: Process, Benefits, Pitfalls	Smartboard, Presentations	Discussion on Benchmarking	https://youtu.be/cS-DNDAPeGI	2	R2
Quiz 2						

Unit 3						
31	Quality function deployment (QFD)	Smartboard, Presentations	Discussion on QFD	https://www.youtube.com/watch?v=u9bvzE5Qhjk	3	R1
Assignment 2						
32	QFD Methodology, Process, Benefits	Smartboard, Presentations	Discussion on QFD	https://youtu.be/iRMsd-X_e-0	3	R2
33	QFD Case Studies	Smartboard, Presentations	Discussion on QFD case study	https://etd.ohiolink.edu/apexprod/rws_etd/send_file/send?accession=ohiou1250171415&disposition=inline	3	R2
34	Taguchi Quality Loss Function: Concept	Smartboard, Presentations	Discussion on Taguchi loss function and problem solving	https://www.youtube.com/watch?v=8u7bTbB1sEQ	3	R2
35	Taguchi‟s Approach Vs Traditional Approach	Smartboard, Presentations	Comparison between Traditional vs Taguchi method	https://espace.curtin.edu.au/bitstream/handle/20.500.11937/51767/250983.pdf?sequence=2&isAllowed=y	3	R2
36	Total Productive Maintenance (TPM): Concept, Objectives	Smartboard, Presentations	Discussion on TPM	http://www.plant-maintenance.com/articles/tpm_intro.shtml	3	R2
37	Twelve Steps for TPM	Smartboard, Presentations	Discussion on TPM	https://www.nikunjbhoraniya.com/2021/04/12-steps-of-tpm-implementation.html	3	R2
38	FMEA	Smartboard, Presentations	Discussion on FMEA	https://youtu.be/IckvY3PA59g	3	R1

39	Need for quality systems	Smartboard, Presentations	Discussion on quality systems need	https://youtu.be/sXhPbXBSU_4	4	T1
40	ISO 9000:2000	Smartboard, Presentations	Discussion on ISO 9000:2000	https://www.adhesivesmag.com/articles/83422-iso-9000-2000-the-new-quality-standard	4	T1
41	Elements of quality systems (such as ISO 9000:2000)	Smartboard, Presentations	Discussion on ISO 9000:2000	https://www.youtube.com/watch?v=gKyecFOnFVI	4	R1
42	Implementation of quality system	Smartboard, Presentations	Discussion on quality system implementation	https://www.youtube.com/watch?v=7jKUNrekqxs	4	R2
43	Documentation in Quality system	Smartboard, Presentations	Discussion on quality system implementation	https://youtu.be/vza6ITkKOF0	4	T1
44	Quality auditing	Smartboard, Presentations	Discussion on quality auditing	https://www.youtube.com/watch?v=fckZK6l678E	4	T1, R2
45	QS 9000 - ISO 14000	Smartboard, Presentations	Discussion on QS 9000 - ISO 14000	https://indiafreenotes.com/iso-9000-qs-9000/ https://www.investopedia.com/terms/i/iso-14000.asp	4	R1
46	Case Studies	Smartboard, Presentations	Discussion on case studies in TQM	https://www.europeanceo.com/business-and-management/total-quality-management-three-case-studies-from-around-the-world/	1,2,3,4	T1, R2

Faculty **Course Mentor** **Chairperson/ Program Head**